Alhassan Alfarran

Python / Full-Stack Developer (FastAPI, React + TypeScript)

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Open to offers · on-site, hybrid or remote · full-time or contract

PROFESSIONAL SUMMARY

Building software commercially since 2024. In 2026 I built semantic and full-text search over technical documentation (FastAPI, sentence-transformers, FAISS, a custom inverted index): an archive of ~4.1k files (~1.5k documents with text), 0.18 s p95 response time, 739 automated tests, accepted by the client. In 2025 I was the lead developer of an internal web system for a network of 9 vehicle depots (React + TypeScript, Node.js/Express, MongoDB, Docker Compose). I develop with AI agents (Claude Code, Cursor, Codex): I set the task, review the changes, check them with tests and on real data, and deploy them.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript

Backend: FastAPI, Node.js / Express, REST APIs, JWT authentication, background jobs, file uploads (Multer)

Search & documents: sentence-transformers, FAISS, inverted index, hybrid search; pypdf, python-docx, openpyxl; OCR (Tesseract, Poppler)

LLM (prototypes built with AI agents): RAG (FAISS + BM25, RRF), OpenAI-compatible APIs, MCP server

Frontend: React 18, TypeScript, Vite, Next.js, Tailwind CSS, responsive UI, framework-free HTML/CSS/JS

Databases: MongoDB (Mongoose); SQLite / PostgreSQL — university project (built with Codex)

Infrastructure & quality: Linux (Ubuntu, systemd), Docker / Docker Compose (basic), GitHub Actions, Git, pytest

Data: pandas, NumPy, scikit-learn — university projects

PROFESSIONAL EXPERIENCE

Python Developer — Elektroservis (Электросервис) Apr 2026 — Oct 2026

Contractor (civil-law contracts) · remote

- Built a search service for an engineering-documentation (PTO) archive: an engineer enters a cable mark, document code, equipment designation or a phrase and gets the document, the page, a text fragment and the network path.
- Hybrid search with no LLM generation: a custom inverted index plus FAISS semantic search on sentence-transformers embeddings; FastAPI backend, plain HTML/JS UI. PDF, DOCX, XLSX and RTF indexing (~4.1k files, ~1.5k with text); OCR of selected scanned pages (Tesseract + Poppler).
- Speed: code and mark queries from 9–10 s to 1.2–1.7 s, phrase queries from 55–70 s to 2–4 s; **0.18 s p95** after moving to the inverted index.
- Quality: **739** pytest tests and a 104-query control set (top-1/top-5); systemd service on Ubuntu Server with atomic index swap and rollback. Accepted by the client in September 2026.
- Built with AI agents (Codex, Cursor, Claude Code): gathered requirements, wrote the agent briefs, reviewed the changes and verified them on the production index.

Full-Stack Developer (lead developer on the project) — Avenue Group May 2025 — Dec 2025
Moscow, Russia · team of 3

- Designed the architecture and API of OKKP, an internal quality-control web system for 9 vehicle depots, and wrote most of the code (~177 of 183 commits in 2025).
- Violations and fines module with server-side filtering and pagination; modules for employees, driver call-backs, a document archive (Cloudinary with local-disk fallback) and notifications.
- Per-depot KPI tiles and in-browser Excel export (SheetJS); JWT authentication (bcrypt); responsive Tailwind CSS UI.
- Deployment with Docker Compose (MongoDB + API) on a Linux server; CI on GitHub Actions. Stack: React 18, TypeScript, Vite, Node.js, Express 5, MongoDB.

Full-Stack Web Developer — Junzi Tech Solutions Oct 2024 — Jul 2025

- Full-stack features for web applications with Node.js, Express, MongoDB and React.
- Cut the response time of a critical API endpoint by **~60%** (caching and a redesigned database query).

Freelance Developer 2024 — Present
Remote

- React / Next.js / TypeScript and Tailwind CSS web applications for clients.

PROJECTS

WhyDoc — prototype, built with Claude Code Oct 2026

- RAG over documents: answers cite the PDF page or Excel cell, and claims are checked against the source. FAISS + BM25 (RRF), FastAPI, LLM via an OpenAI-compatible API.

Lumen Search — open-source prototype, built with AI agents Oct 2026
github.com/kyan9400/lumen-search · live demo: kyan9400.github.io/lumen-search

- Hybrid search with citations over PDF, Word, Excel and scanned pages in English and Russian: BM25 + multilingual embeddings in FAISS, Tesseract OCR, FastAPI.

Pathwise — master's thesis, built with Codex 2026

- Recommender system for study programmes: FastAPI, scikit-learn (TF-IDF + cosine similarity), pandas, PostgreSQL or SQLite; Next.js front end.

EDUCATION

Master's Degree — National University of Science and Technology MISIS, Moscow 2024 — 2026
Computer and Information Sciences

Bachelor's Degree — Ural Federal University, Yekaterinburg 2020 — 2024
Computer Science

CERTIFICATIONS (SELECTED)

- Full-Stack Web Development with Angular — Coursera, HKUST
- Server-side Development with Node.js, Express and MongoDB — Coursera
- Introduction to SQL; Understanding and Visualizing Data with Python — Coursera, University of Michigan
- Machine Learning for All — Coursera, University of London
- Programming with Google Go Specialization — Coursera
- JavaScript Algorithms and Data Structures; Data Visualization — freeCodeCamp
- ICDL — International Computer Driving Licence

LANGUAGES

Arabic — native · **English** — C1 (professional) · **Russian** — B2 (professional working)